Health & Safety

Standard Hot works

Reference

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A	01-11-2023	O.BRZEZINSKI	I.LELIEVRE	L.KNOLL	First issue
B	21-05-2024	A.JANSSENS	I.LELIEVRE O.BRZEZINSKI	L.KNOLL	Modification of 1. hot works

The standards presented in this document serve as a practical guide to define the means to be planned and deployed to effectively manage and conduct hot works.

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1. Hot works

"Hot work" refers to all work likely to produce a source of ignition (heating, sparks, flames, etc.)

It is necessary for such operations :

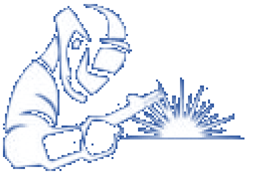
- To use compliant equipments that is in perfect working conditions (torch, welding stations, grinders, etc.)
- To set up a dedicated fire extinguisher in the immediate vicinity of the workstation
- To ensure an annual check of fire extinguishers, properly registered and recorded

It is recommended to implement a 2-hour monitoring period after hot work to ensure the absence of fire initiation.

Fire extinguishers will under NO CIRCUMSTANCES be used for gas fires (specific instructions)

2. Operator competencies

Depending on the types of hot works to be carried out (welding, oxy-fuel cutting, grinding, etc.), different qualifications are required.
Contact Health & Safety Department if necessary.



3. Fire permit / Hot works permit

A “Hot works permit” may be deemed compulsory, in particular (non-exhaustive list):

- For hot work in tunnel
- On a tunnel boring machine
- In confined spaces or areas not naturally ventilated
- With private clients who impose it in their standards (industries)
- Etc.

Safety risks analysis will determine whether or not it is necessary to put in place a “Hot work Permit” procedure.

Fire license stump book

4. Conditions of use of gas cylinders

Check the gas pipes before each use :

- Cracks, dirt, expiry date (5 years)
- No screwed hoseclamp on the gas tubes
- Flame arrestor check valves at the outlet of the regulator and as close as possible to the torch
- Have fire extinguisher available near the place of use
- Have a clean and tidy work area, without potentially flammable equipment or materials nearby

Deteriorated pipe

Reduction of arduousness = use of rolling cart

Flame arrestor check valve

Regulator

Wear specific PPE according to the tasks (welding, oxy-fuel cutting)

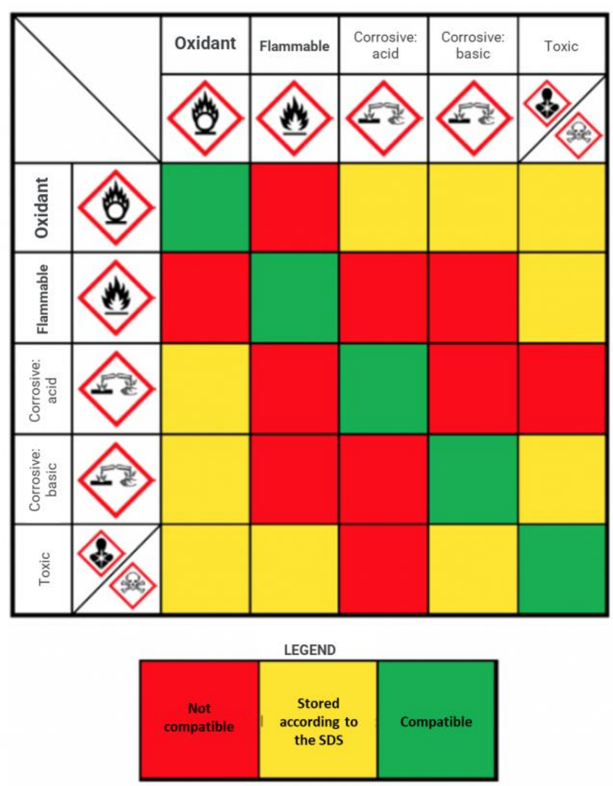
5. Storage conditions for gas cylinders

- The cylinders are correctly labeled, otherwise their use is prohibited and they must be removed from the site
- Store cylinders upright and strapped to avoid tipping inside the rack
- A minimum distance of 3 meters is ensured between full cylinders and empty cylinders of the same gas
- Flammable cylinders must be stored at a good distance from oxidizer bottles

Prohibition on connecting the torch directly into the rack with several bottles

- Protect gas cylinders from the sun when stored in the open air – Covers should not be in direct contact with the cylinders
- SDS are available on site and consulted to ensure compatibility of stored products
- It is prohibited to store gas cylinders in material containers
- It is prohibited to store a gas bottle near the air intake of a compressor
- It is prohibited to store gas cylinders in a confined space
- It is prohibited to store gas cylinders under pressure with other chemical products.
- Limit the quantity on site as much as possible

- Any damaged or overheated gas cylinder must be placed in quarantine and labelled as prohibited for use – Contact the supplier for the procedure to follow
- The storage racks can be slinged from the ground
- Do not lift gas cylinders by their caps



6. Welding works

Material :

- Use compliant equipment that is in perfect working condision
- Do not disassemble the tools for home-made repair
- The grounding must be connected to the part to be welded and not on its support

Any defective welding equipment is :
- Removed from production areas and stored under lock and key
- Prohibited to be used and identified as such by a sticker or other visual means as defined in site procedures

Collective protections :

- Aspiration directly at the indoor source and sufficient ventilation of premises, boxes, etc.
  (See Section 7 "Welding fumes")
- Limit coactivity and access to welding areas
- Use of a protective screen in case of passing nearby or coactivity
- Extinguishing means near the workstation

No hot work near dangerous chemicals (flammable, oxidizing, etc.)

Source aspiration

Protective screen

Specific individual protections :

- Heat-resistant protective gloves with cuffs
- Leather apron
- 2-in-1 ventilated welding helmet : visual and respiratory protection
- Gaiter if safety shoes with laces
- Cotton (non-flammable) and clean clothing
- No retroreflective vest not intended for flame protection

2-in-1 mask

Flame retardant clothing

Gaiters

Apron

Welder gloves

7. Welding fumes

The fumes, mixed with hot air, are formed of particles and gases capable of reaching the alveolar region of the respiratory system.
Depending on their compounds, fumes can cause irritation, respiratory diseases and potentially cancer.

		WORKSHOP	CONSTRUCTION SITE	CONFINED SPACE
COLLECTIVE PROTECTIONS	Source aspiration	By decreasing priority: - Suction integrated into the tools (suction torch) air flow 100 m ^{3}/h - Suction template air flow 50 to 150 m ^{3}/h - Suction table air speed at the borders 0.5m/s - Welding cabin air speed in opening 0.5m/s - Articulated arm air speed at the welding point 0.5 m/s -General ventilation: *by principle, general ventilation alone is not sufficient as a means of prevention.*	- Mobile autonomous suction arm 	- Suction integrated into the tools (suction torch) air flow 100 m ^{3}/h - Autonomous suction arm 
	Substitution	- Welding with cored wire to be replaced with welding under protective gas with a less emissive solid wire - Favor processes without filler metal or with low-emission filler metal (sub-flux welding, semi-auto TIG, pulsed MIG, etc.) - Reduce current intensity and electrode diameter, arc length - Choose the least toxic and least emissive welding rods - Solvent cleaning (CHLORINE-FREE) and drying time		
	General mechanical ventilation	***Calculated according to the useful volume of the room or building to be ventilated and the efficiency of the suction devices at the source*** - Dilute residual pollutants - Air flow of 60 m ^{3}/h per occupant - Provide an introduction of fresh air of an equivalent volume to heat in cold periods (opening doors should be avoided)	- Stand back from the weld pool - Position yourself in the direction of the wind to avoid to inhale the fumes	***- Minimum 20 volumes of air inlet per hour and more if the quantity of contaminant is higher.*** - Stand back from the weld pool
INDIVIDUAL PROTECTIONS		**In addition to a collective protection system and for outdoor works :** **-** Powered ventilation mask with P3 anti-aerosol particle filter + ABEK type anti-gas filter.  3M SPEEDGLASS 9100 FX mask With double visor		- Insulating respiratory protection device with uncontaminated air supply.  3M SPEEDGLASS 9100 FX Air ADFLO mask

8. Oxycutting work – 1/2

The use of torches or oxycutters for cutting metal elements involves risks linked to the pressure of gases and their combustion, projections, flames, radiation and smoke.
Compliant equipment should be used and in good conditions.

Preparing the workstation :

- Workstation is ventilated and away from storage areas and pedestrian walkways
- Block/retain the piece to be cut to limit falling objects (non-flammable cushions)
- Do not place cables or flammable materials in the area of incandescent projections
- Provide suitable extinguishing means
- Check the condition and conformity of any equipment before using it
- Source suction or assisted ventilation mask in enclosed or confined spaces

Storage and movement :

Specific instructions (non-exhaustive list):

- Store gas cylinders upright and properly secured in a rack or handling cart
- Bottle lying = potential gas leak (liquid)
- Transport rack to be used only to tranfer the gas cylinders
- Storage on wheeled cart when on use

Specific PPE:

Welder's jacket or flame-retardant clothing

Welding or flame retardant gloves

Lampworker's glasses with shade 5 to 8

Life jacket or harness and fire-retardant accessories if necessary

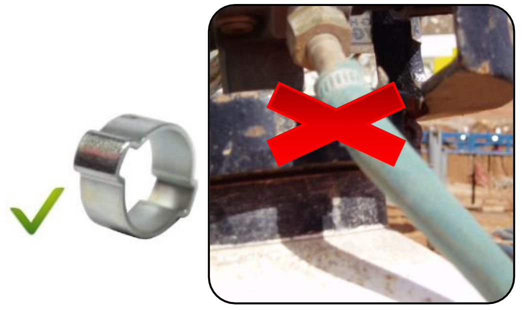
8. Oxycutting work – 2/2

Compliance and control of equipment :

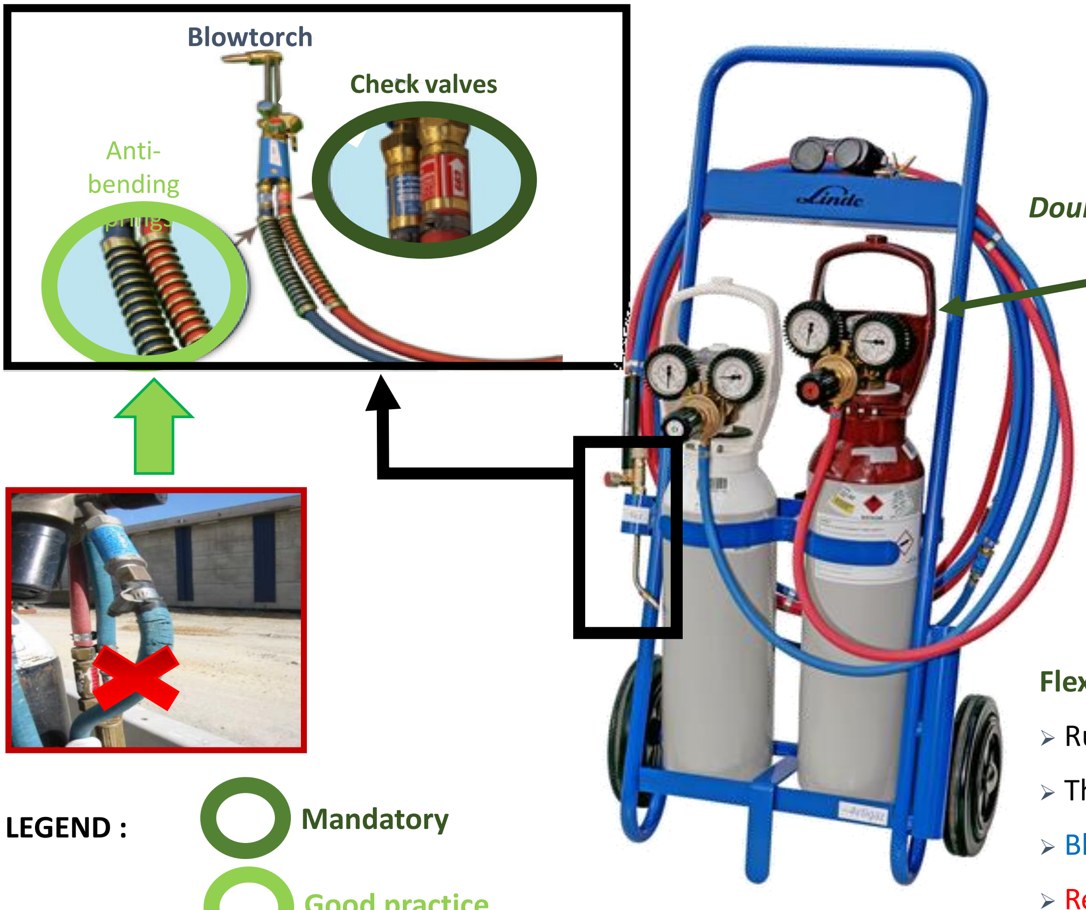
Before any use, it is mandatory to check:

- Cylinders and regulators
- Flexibles
- Connections and check valves
- Blowtorch

Mandatory check valves on the torches and at the outlet of each regulator



Exclusive use of ear clamps / SERFLEX prohibited to secure the flexibles



Flexible pipes :

- Rubber EN 559
- Thermoplastic EN 1327
- Blue: oxygen
- Red: propane/acetylene

LEGEND :

- **Mandatory**
- **Good practice**

9. Plasma cutting

Workstation preparation:

- Workstation ventilated and **away from storage areas and pedestrian walkways**
- **Block/retain the piece to be cut** to limit falling objects
- **Non-flammable blocks** are placed under the piece to be cut:
  - Near the edge (at each end of the piece)
  - On either side of the cut line
- Ensure a security perimeter, protect with blackout curtains
- Be careful not to place cables or flammable objects in the area of incandescent projections
- Provide extinguishing means
- Provide compressed air and electricity supplies and secure air flows with strops
- Provide welding glasses for other workers nearvy, in case of coactivity
- Source suction or assisted ventilation mask in enclosed or confined spaces

Before the use of the equipment:
- Check the condition of the lances and flexibles
- Change the assembly every 200 hours of cutting
- Check the grounding cable of the workpiece

Protective screen

Use of a compressed air compressor = respect the safety rules for using this type of equipment

Specific PPE:

Welder's jacket or flame-retardant clothing

Welding or flame retardant gloves

Lampworker's glasses with shade 5 to 8

Hearing protection adapted to noise with compressed air